Solar Plant Manager

Monitoring, forecasting and fault detection for small solar plants

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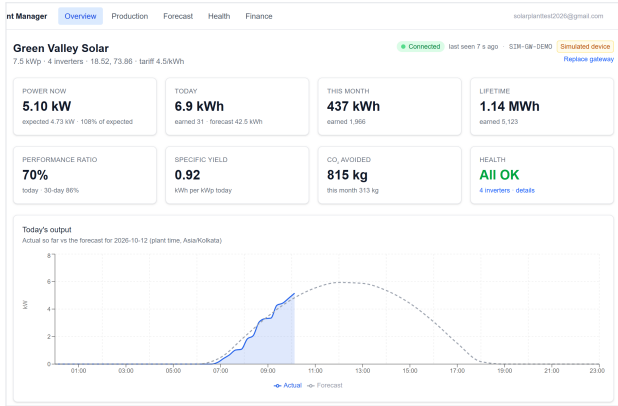
Data Analytics 2026

What we built

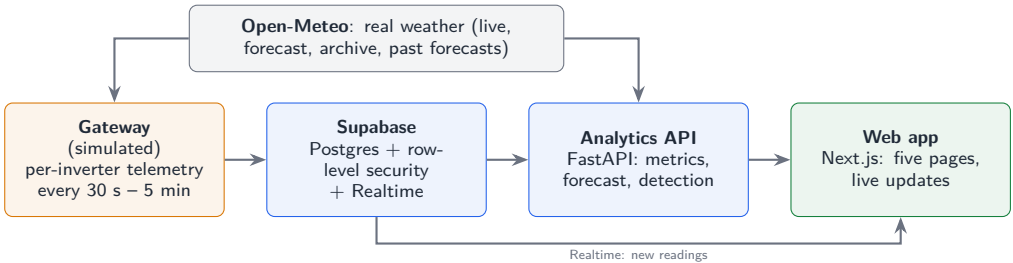
A day-to-day manager for a small plant's owner:

- **Overview**: power now vs expected, energy, earnings, CO_2, performance ratio
- **Production**: daily and monthly yield, inverter comparison
- **Forecast**: next 7 days, with measured accuracy
- **Health**: live fault detection, alerts, where energy is lost
- **Finance**: projected year, payback

Sign up, describe the plant, pair its gateway with a code; updates live.



How it works



The data

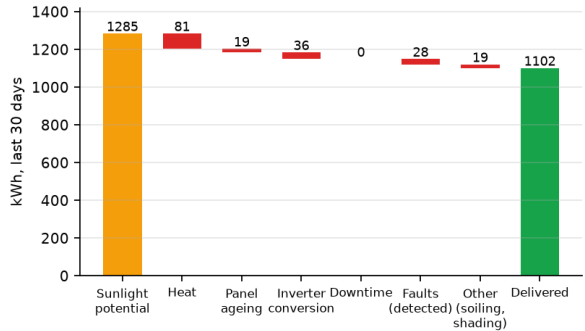
- **Inverters are simulated** (no hardware): **real weather** → pvlib physics (sun angle, tilt, heat, ageing)
- Real-life mess added: 3% sensor noise, ±6% sky error, shading, faults of 5–40% for 2–8 h
- Every fault is logged: **an answer key to measure detection**

The system

- The gateway pairs with a code and holds its own device token, like a real data logger
- The API reads data *as the signed-in user*: no admin keys

Plant metrics

- **Performance ratio** (IEC 61724) = energy ÷ (sunlight on panels × kWp). Healthy: 75–85%; test plant: 86% (30 d)
- **Specific yield** = kWh per installed kWp, to compare plants of any size
- **Availability**, earnings, CO_2 avoided (0.716 kg/kWh, Indian grid)
- **Loss breakdown**: sunlight potential → heat → ageing → inverter → downtime → faults → delivered (adds up exactly)
- **Payback**: a typical year of real weather at the site: 13.8 MWh, 5.6 years on 3.5 lakh



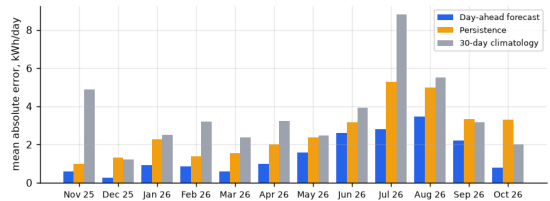
Where the test plant's energy went, last 30 days (Health page).

Forecasting: does it beat the naive guess?

Weather forecast → physics → × plant calibration (actual ÷ physics, last 14 days). Scored against the forecast *as issued the day before*.

12 months, Pune	Error (kWh/day)	Forecast is
Day-ahead forecast	**1.54** (4.6%)	
Same as yesterday	2.65	**42% better**
30-day average	3.72	**59% better**
Monsoon (Jun–Sep)	2.77 (8.9%)	34% better

In the app: the last 7 days' accuracy (±5.7%), 35% better than "same as yesterday".



Ahead of both naive methods in every month; the monsoon triples the error.

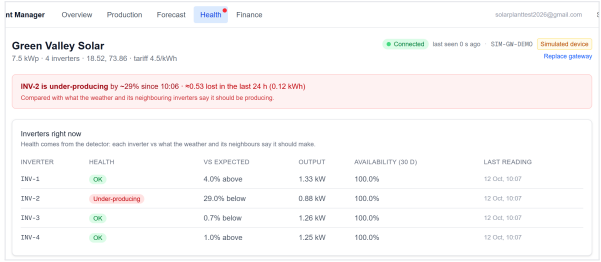
Fault detection

Learn what healthy output looks like, flag the shortfall

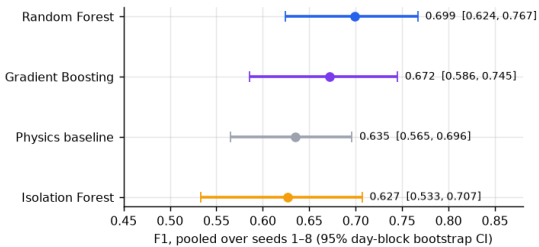
- Expected power from weather + **the median of the other inverters**
- Shortfall = (expected − actual) ÷ expected, smoothed
- Threshold = median + 2.5 robust σ of held-out residuals: **never uses the labels**

Four detectors compared: physics baseline, Random Forest, Gradient Boosting, Isolation Forest (unsupervised)

Live: a 30% fault injected on the gateway is flagged within 1–3 minutes.



Which detector? Only some differences are real

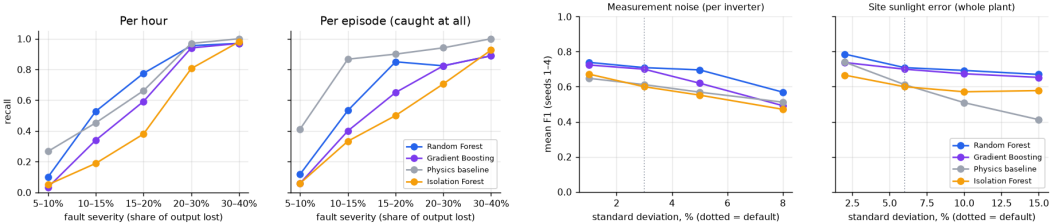


F1 pooled over 8 seeds; 95% CI by day-block bootstrap.

- Random Forest **beats** the physics baseline (+0.065, CI +0.007 to +0.124) and Isolation Forest
- Random Forest vs Gradient Boosting: **tied** (CI crosses 0)
- Per episode, a trade-off per-hour F1 hid:

	Episodes caught	Alarms that are real
Random Forest	68%	65%
Gradient Boosting	60%	72%
Physics baseline	**84%**	35%

Detection limit, and what makes the ML worth it



- **The limit as a curve**: Random Forest catches 10% of hours at 5–10% loss, 53% at 10–15%, 96%+ above 20%. Mild faults hide in normal noise.
- **Sky error separates the methods**: as the plant-wide sunlight error grows 2% $\rightarrow$ 15%, the physics rule falls 0.74 $\rightarrow$ 0.41 F1; Random Forest only 0.79 $\rightarrow$ 0.67.
- **Why**: the Random Forest is effectively a **peer comparison**. Shuffling the neighbours' output drops R^2 by 1.5; any weather input by <0.02. Neighbours share the sky error, so it cancels. Blind spot: a fault hitting *every* inverter at once.

Limitations and next steps

Limitations

- Inverter output is simulated; detection numbers describe this simulator (and we showed how they move with it)
- Benchmark: one month, one site; the forecast study covers a full year
- Forecast "actual" is reanalysis weather, not an on-site sensor
- Model settings tuned on the seeds they're scored on (mild optimism)

Next

- Real data in: CSV import from an inverter portal, then real hardware
- Peer comparison + weather check as the detector (F1 0.78 in experiments)
- Soiling and cleaning advice; email alerts
- Deployment

Takeaways

42%
lower forecast error
than "same as yesterday"

0.70
F1 fault detection
(CI 0.62–0.77)

>20%
faults caught
~96% of the time

5.6 yr
payback, from a
typical year of weather

- A working plant manager: live monitoring, metrics, forecasts, health and finance on one plant's own data
- Simulated faults on real weather made detection *measurable*; bootstrap CIs show which differences are real
- ML earns its place where the weather data can't be trusted: comparing inverters with each other

Thank you Questions?